Scientific Method - Controls and Variables

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Definition of Key Terms

- ***Control*** - A part of the experiment that is not being tested and is used for comparison.
- ***Variable*** - Any part of an experiment that can change.
- ***Independent Variable*** - The part of the experiment that is changed by the scientists or person performing the experiment.
- ***Dependent Variable*** - The part of the experiment that is affected by the independent variable.

Read the description for each experiment and answer the questions.

Krusty Krabs Breath Mints

Mr. Krabs created a secret ingredient for a breath mint that he thinks will "cure" the bad breath people get from eating crabby patties at the Krusty Krab. He asked 100 customers with a history of bad breath to try his new breath mint. He had fifty customers (Group A) eat a breath mint after they finished eating a crabby patty. The other fifty (Group B) also received a breath mint after they finished the sandwich; however, it was just a regular breath mint and did not have the secret ingredient. Both groups were told that they were getting the breath mint that would cure their bad breath. Two hours after eating the crabby patties, thirty customers in Group A and ten customers in Group B reported having better breath than they normally had after eating crabby patties.

1. Which people are in the control group?
   a. Group A
   b. Group B

2. What is the independent variable?
   a. The actual breath mint.
   b. The crabby patties.
   c. The secret ingredient in the breath mint.
   d. How many crabby patties eaten.

3. What should Mr. Krabs' conclusion be?
   a. The breath mint with the secret ingredient does reduce breath odor.
   b. The breath mint with the secret ingredient reduces breath odor over 50% of the time.
   c. The breath works, but it is not 100% effective.
   d. All of the above.

SpongeBob Clean Pants

SpongeBob noticed that his favorite pants were not as clean as they used to be. His friend Sandy told him that he should try using Clean-O detergent, a new laundry detergent she found at Sail-Mart.

SpongeBob made sure to wash one pair of pants in plain water and another pair in water with the Clean-O detergent. After washing both pairs of pants a total of three times, the pants washed in the Clean-O detergent did not appear to be any cleaner than the pants washed in plain water.

4. What was the problem SpongeBob wanted to investigate?
   a. Is Clean-O detergent effective?
   b. Is the length of time the pants are washed important?
   c. How does water temperature affect cleaning pants?
   d. Does how often I wash my pants affect how clean they are?

5. What is the independent variable?

a. Water temperature. b. Length of wash time. c. Laundry soap d. Size of washing tub

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6. What should Sponge Bob's conclusion be?
a. Clean-O best cleans his pants.
b. Plain water best cleans his pants.
c. Cold water best cleans his pants.
d. Clean-O is not effective cleaning his pants.

Squidward's Symphony

Squidward loves playing his clarinet and believes it attracts more jellyfish than any other instrument he has played. In order to test his hypothesis, Squidward played a song on his clarinet for a total of 5 minutes and counted the number of jellyfish he saw in his front yard. He played the song a total of three times on his clarinet and repeated the experiment using a flute and a guitar. He also recorded the number of jellyfish he observed when he was not playing an instrument. The results are shown in the chart.

Number of Jellyfish/Instrument				
Trial	***No Music***	***Clarinet***	***Flute***	***Guitar***
1	5	15	5	12
2	3	10	8	18
3	2	12	9	7

7. What is the independent variable?
a. Number of jellyfish. c. Length the music was played.
b. Instrument used. d. The song he played.

8. What should Squidward's conclusion be?
a. The clarinet and guitar attracted the same number of jellyfish.
b. The flute attracted more fish than the control (no music).
c. Music attracts more jellyfish than does no music.
d. All of the above.

Super Bubbles

Patrick and SpongeBob love to blow bubbles! Patrick found some Super Bubble Soap at Sail-Mart. The ads claim that Super Bubble Soap will produce bubbles that are twice as big as bubbles made with regular bubble soap. Patrick and SpongeBob made up two samples of bubble solution. One sample was made with 5 oz. of Super Bubble Soap and 5 oz. of water, while the other was made with the same amount of water and 5 oz. of regular bubble soap. Patrick and SpongeBob used their favorite bubble wands to blow 10 different bubbles and did their best to measure the diameter of each one. The results are shown in the chart

***Bubbles* (Diameter in centimeters)**		
Bubble	***Super Bubble***	***Regular Soap***
1	15	10
2	10	5
3	12	16
4	18	14
5	22	11
6	13	12
7	16	11
8	18	15
9	15	15

9. What did the Super Bubble ads claim?
a. Super Bubble produces bubbles twice as large as regular bubble soap.
b. Super Bubble makes twice as many bubbles as regular bubble soap.
c. Super Bubble bubbles last 2x long as bubbles from regular bubble soap.
d. Super Bubble is cheaper than regular bubble soap.

10. What is the independent variable?
a. Bubble size. c. Length of time bubbles last.

b. Number of bubbles made.
d. Type of bubble solution.

11. What should their conclusion be?
    a. Super Bubble solution did not produce bubbles twice as large as those made with regular bubble soap.
    b. Regular bubble soap lasts twice as long as Super Bubble.
    c. Bubbles made with Super Bubble last twice as long as bubbles made with regular bubble soap.
    d. There was no difference between Super Bubble and regular bubble soap.

Slimotosis

Sponge Bob notices that his pal Gary is suffering from slimotosis, which occurs when the shell develops a nasty slime and gives off a horrible odor. His friend Patrick tells him that rubbing seaweed on the shell is the perfect cure, while Sandy says that drinking Dr. Kelp will be a better cure. Sponge Bob decides to test this cure by rubbing Gary with seaweed for 1 week and having him drink Dr. Kelp. After a week of treatment, the slime is gone and Gary's shell smells better.

12. What mistake was made in this experiment?